

WASP-39b

R-1697

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-39_140517 · created 2026-08-02T13:49:22+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2456794.754 +/- 0.047 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.19 +/- 0.053
Transit depth (Rp/Rs)^2	3.61 +/- 2.01 [%]
Orbital Inclination (inc)	85.88 +/- 2.11
Airmass coefficient 1 (a1)	0.956 +/- 0.051
Airmass coefficient 2 (a2)	0.01 +/- 0.12
Scatter in the residuals of the lightcurve fit is	5.33 %
Best Comparison Star	#3 - [119, 361]
Optimal Aperture	2.7
Optimal Annulus	11.57
Transit Duration (day)	0.064 +/- 0.042

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.19 ± 0.053 vs 0.1457 (deviation 0.84σ)
Duration fit vs published	0.064 d vs 0.12 d (deviation 1.33σ)
Mid-time O–C	-8.5 min
Depth SNR	3.6
Residual scatter	5.33 %

Light curve

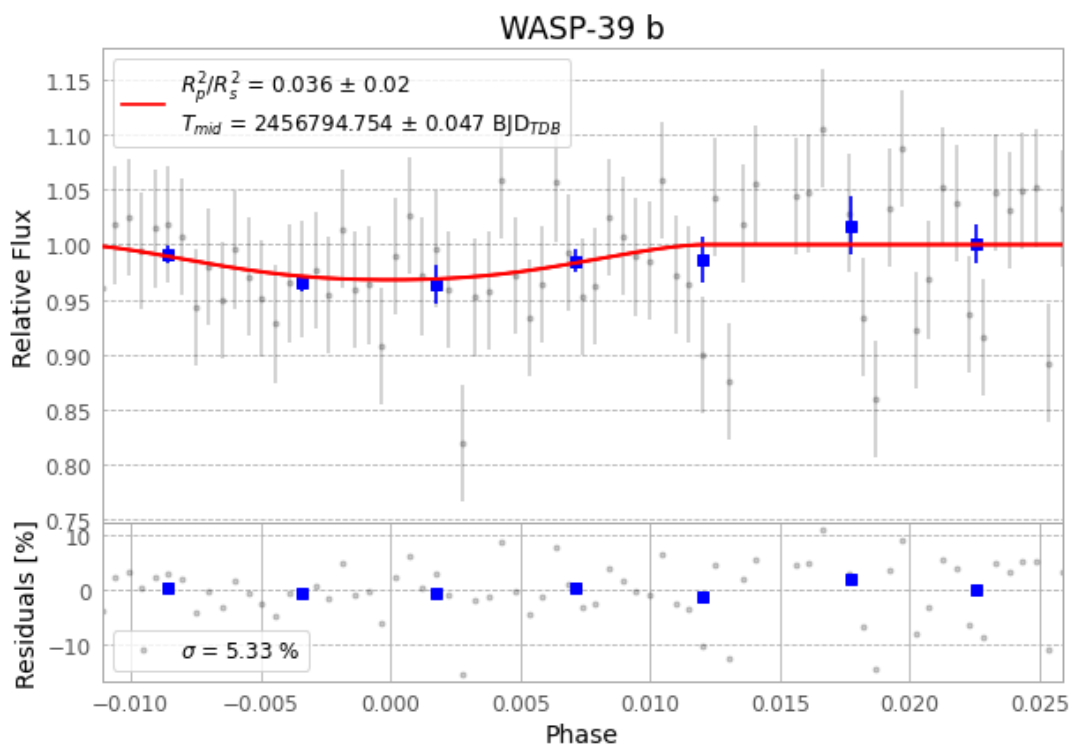


Figure 1 — FinalLightCurve_WASP-39 b_2014-05-16.png (SHA-256 d35d79114fe2872b...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [184, 224], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 fe91380a7d6f0acd...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

73 FITS frames (48 MB), WASP-39140517045113.FITS → WASP-39140517082712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-39-140517.log (b2aeedc4ee49...), FinalLightCurve_WASP-39 b_2014-05-16.png (d35d79114fe2...), FinalLightCurve_WASP-39 b_2014-05-16.csv (8684d32aa844...)

Compute resources

Wall time 175.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 13:49Z.